

Documentation of Academic AI Assistance

MA289: Intro to Statistical Learning • Project 3: Applied Demonstration of Regularization • West Point, NY • 22 APR 2026

STUDENT

CDT Zander Bos | Instructor: MAJ Patrick Kuiper | Course: MA289

AI TOOL USED

Claude (Anthropic). Assistance given to the author, CDT Zander Bos. Accessed via claude.ai. Anthropic, (<https://claude.ai>). West Point, NY, 22APR2026.

CONVERSATION TRANSCRIPT

Full transcript of the AI consultation is available at the following link:

<https://claude.ai/share/335d389e-70ff-4fc3-9405-c3dfbb5e0d46>

PROMPTS USED

The following prompts were submitted to Claude across multiple sessions pertaining to MA289 Project 3: "What is a good segue from Assignment 2," "Help me create the code necessary to process the sklearn diabetes dataset and fulfill the requirements for Project 3," "Give me all the blocks but still explain all of them," "I want it to be completely diabetes data," "Give me every block of the project explained," "Yes, and give the completed code to me in chat instead of a file," "I have to present the research - help me transform Project 3 into the following parameters: Problem/Data, Model, Error, Optimization; present a notebook; only real data; code that finds the parameters for the models; compare Ridge and Lasso versus Best Subset (AIC and BIC); only interesting results; standardize lambda and alpha," "I want you to make the part I am presenting a lot shorter so I can give a decent explanation in 10 minutes," "Create a works cited," "Could you create a visualization for the diabetes dataset before we do anything with it," "How are they measuring diabetes progression," "Explain what `ridge_cv = RidgeCV(alphas=lambdas, cv=5).fit(X_train_s, y_train)` does," "What does `max_iter=10000` mean," "Why does Best Subset select different features than Lasso," "What is the problem this project is trying to solve," "What does 5-fold cross-validation do in this context," "Write this in Overleaf code nicely formatted," "Help me understand all of the clinical features," "Is Best Subset considered a model," "What do AIC and BIC stand for," "Tell me the difference," and "What would be a good title for the presentation."

WHAT THE PROMPTS RETURNED

The prompts returned: a restructured eight-cell presentation notebook following the Problem-Model-Error-Optimization framework using only the sklearn diabetes dataset with consistent lambda notation throughout; complete Python code for a Google Colab notebook implementing OLS, Ridge, Lasso, and Best Subset Selection (Algorithm 6.1, ISLR p.231) on both a synthetic sparse dataset and the sklearn diabetes dataset; eight figures including coefficient profiles, regularization paths, actual versus predicted scatter plots, Best Subset AIC/BIC curves, a feature inclusion heatmap, a synthetic versus real comparison, and a full method comparison; two

exploratory data visualizations including a feature correlation heatmap and target distribution scatter plots; a Chicago-style works cited list; LaTeX code for the written report formatted for Overleaf; conceptual and line-by-line explanations of all Python code including cross-validation, model fitting, regularization paths, and the Best Subset algorithm; and explanations of all statistical and clinical concepts including the ten diabetes dataset features, AIC, BIC, MSE, RSS, five-fold cross-validation, regularization strength, and the geometric difference between L1 and L2 penalties.

HOW THE INFORMATION WAS USED

The generated code structure, figure design, and analytical framework informed the construction of the Project 3 Google Colab notebook and presentation. The conceptual explanations were used to deepen understanding of the statistical methods and to inform the written sections of the technical report. Of the code returned, the sklearn implementations, statsmodels AIC/BIC calculations, and mathematical formulations were verified against course materials and documentation prior to submission.

AUTHOR'S OWN WORK

All written analysis, mathematical formulation, verbal presentation delivery, and interpretation of results are the author's own work, informed by but not taken verbatim from these consultations. All writing submitted in the technical report was the author's own. The author used the generated code and explanations as a starting point and reference, not as final submitted work without independent verification and understanding.

COMPLIANCE STATEMENT

This documentation is submitted in accordance with USMA Documentation of Academic Work (DAAW) guidelines for AI-assisted academic work. The full conversation transcript is accessible at the link provided above.
